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3. Magnesium alloys are used in the casings of tablet computers and mobile phones.

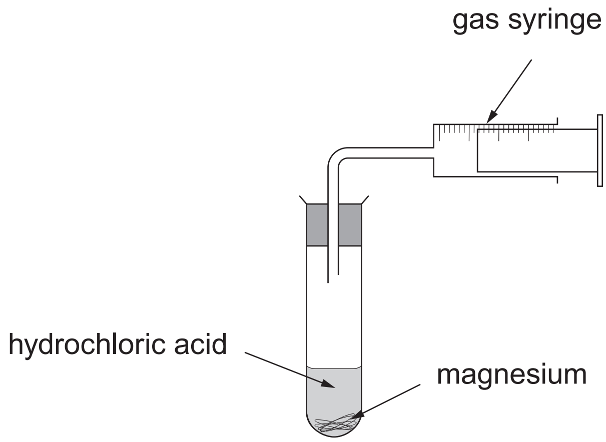
(a) Complete the sentence below by underlining the correct word in the brackets. [2]

An alloy is a **(mixture / compound / solution)** of two or more elements. One of these must be a **(gas / metal / liquid)**.

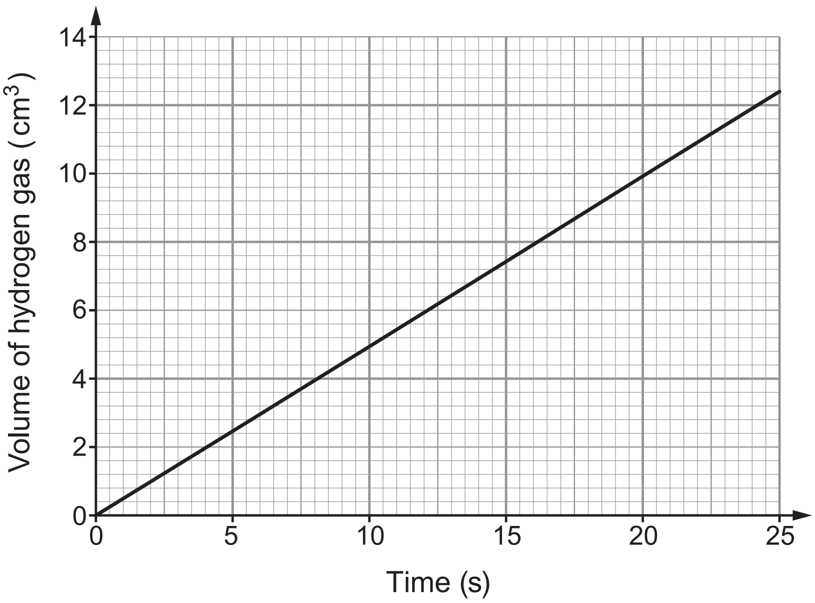
(b) Magnesium reacts with hydrochloric acid producing hydrogen gas.

magnesium + hydrochloric acid $\longrightarrow$ magnesium chloride + hydrogen

A student investigated this reaction using the apparatus below.



A graph of the student's results is shown below.



Draw a line on the graph above to show the results that you would expect if the experiment was repeated with a lower concentration of hydrochloric acid. [1]

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(c) The student also investigated the reaction of hydrochloric acid and magnesium at different initial temperatures.

Initial temperature (°C)	Time taken for magnesium to disappear (s)
10	80
20	40
30	20
40	10
50	5

(i) State **three** variables that need to be controlled in this experiment. [3]

1. ....

2. ....

3. ....

(ii) I. The student thinks that every time the temperature doubles, the time taken halves. Explain whether you agree. [2]

....

....

....

....

II. Predict the time it would take for the magnesium to disappear at 60 °C. [1]

time = ... s

(iii) Complete the following sentences by <u>underlining</u> the correct words in the brackets. [2]

As the temperature increases, particles move **(faster / slower / at the same speed)**.

The particles collide **(more often / as often / less often)** in a given time.

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